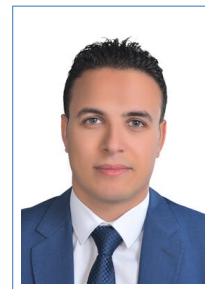


Mohamed Sobhy Elbakry

Curriculum Vitae

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Professional Summary

Distinguished associate professor in electronics and communications engineering (ECE) department at institute of aviation engineering and technology (IAET) – Egyptian aviation academy (EAA). I am adjunct at Mechatronic program, Faculty of engineering, Cairo University and ECE department, higher technological institute (HTI), 10th of Ramadan, and 6th of October branch. Demonstrated expertise in electrical and optical communications, digital signal processing, and mobile communication systems. Strong record of guiding innovative graduation projects, such as IoT processors and autonomous mobile robots, in collaboration with top international research labs. Skilled in supervising competitive teams and contributing to international conferences as a TPC member and session chair.

Published numerous papers in high-impact journals and conferences, focusing on multicarrier systems, interference mitigation, and deep learning for communication systems. These articles are centered on the performance analysis on optimizing spectral efficiency, capacity, and robustness in different environments (wireless, optical, underwater). Looking ahead, my research will explore next generation communication technologies, including 5G, 6G and beyond 6G (B6G), with a focus on massive MIMO, reconfigurable intelligent surface (RIS), and fluid antenna systems (FAS). My focus is performance analysis of communication systems using simulation and mathematical tools; however, I am open for any new related areas. Driven by a commitment to pushing the boundaries of wireless technology, I am eager to contribute to pioneering advancements that will define the future of global communication systems. Proficient in MATLAB, Python, and LabVIEW, with a commitment to advancing the field through cutting-edge research and educational excellence.

Academic Experiences

- 2024 - Present **Associate Professor**, Electronics & Communications Engineering (ECE) Department, Institute of Aviation Engineering & Technology (IAET), Egyptian Aviation Academy (EAA)
- 2018 - 2024 **Assistant Professor**, ECE Department, IAET, EAA
- 2022 - Present **Adjunct Assistant Professor**, Credit Hour Mechatronic Program, Faculty of Engineering, Cairo University.
<http://eng.cu.edu.eg/ar>
- 2020 - Present **Adjunct Assistant Professor**, ECE Department, Higher Technological Institute (HTI), 10th of Ramadan, 6th of October Branch, <http://www.hti.edu.eg>
- 2005 - 2018 **Teaching Assistant**, ECE Department, IAET, EAA <http://iaet-edu.net>

Administrative Experiences

- 2024 - Present Team member of the board of the EAA
- 2021 - Present Head of the department of ECE Department, IAET, EAA
- 2021 - Present Head of the control board, IAET, EAA
- 2021 - Present Team member for preparation the file for engineering public sector committee, IAET, EAA
- 2020 - Present Head of the advanced communication LAB, ECE Department, IAET, EAA
- 2015 - Present Team member in the quality assurance unit, ECE Department, IAET, EAA
- 2021 - 2022 Head of the department of information technology (IT), IAET, EAA
- 2018 - 2021 Team member in the control board and questions, IAET, EAA
- 2015 - 2016 Team member for experiments preparation on advanced communication LAB, ECE Department, IAET, EAA

Volunteer Experiences

- 2024 Member of organization committee for a seminar entitled "Network & Mobile Communications", Senior Planning and Optimizer Engineer
- 2023 Member of organization committee for a seminar entitled "Services of Information Technology Institute (ITI), ITI CREATIVA - Cairo University
- 2022 Member of organization committee for a workshop entitled "NVIDIA Deep Learning", NVIDIA company
- 2022 Member of organization committee for a seminar entitled "Mobile Communications & Embedded Systems Careers", National of Telecommunication Institute (NTI) & Senior Planning and Optimizer Engineer
- 2022 Member of organization committee for a seminar entitled "Space Day Event", Egyptian Space Agency (EgSA)

Teaching Experiences

Throughout my academic career, I have gained extensive teaching experience in the field of Electronics and Communications Engineering, serving as both a Teaching Assistant, an Assistant Professor and Associate Professor. My teaching roles have encompassed a wide range of subjects, from foundational courses to advanced, specialized topics.

- 1. Electrical Communications (1-A), **(ECE 303)**, **IAET**, **HTI**
- 2. Electrical Communications (1-B), **(ECE 307)**, **IAET**, **HTI**
- 3. Electrical Communications (2-A), **(ECE 403)**, **IAET**, **HTI**
- 4. Electrical Communications (2-B), **(ECE 407)**, **IAET**, **HTI**
- 5. Optical Communications, **(EEC 242)**, **HTI**
- 6. Signal Analysis (A), **(ECE 203)**, **IAET**, **Cairo University**
- 7. Signal Analysis (B), **(ECE 206)**, **IAET**, **Cairo University**
- 8. Digital Signal Processing (DSP), **(ECE 312)**, **IAET**
- 9. Mathematics (3-A), **(BAS 201)**, **IAET**
- 10. Mobile Communication Systems, **(ECE 410)**, **IAET**
- 11. Satellite Communication Systems, **(ECE 417)**, **IAET**
- 12. Circuits (1), **(ECE 101)**, **IAET**, **HTI**
- 13. Circuits (2), **(ECE 103)**, **IAET**, **HTI**
- 14. Active Circuits, **(ECE 303)**, **IAET**, **HTI**
- 15. Computer Networks, **(EEC 210)**, **HTI**
- 16. Computer Skills, **(CSC 001)**, **HTI**

Graduation Projects

- 2024 "Design, Implementation and verification of Low-Power IoT RISC-V Processor with Hybrid Encryption Accelerator", this project was cooperation with NISC, Nile University, Giza, Egypt through Prof. Ahmed Madian
- 2023 "Digital Twin Simulation of Connected and Automated Vehicles with the Unity Game Engine", this project was fully funded from Academy of Scientific Research and Technology (ASRT), Cairo, Egypt
- 2022 "Autonomous Navigation for Indoor Mobile Robot: Hardware Implementation", this project was cooperation with Robotics LAB, Indiana University, USA through Dr. Ihab S. Mohamed
- 2020 "Autonomous Mobile Robot Navigation in Dynamic Environments: Deep Learning-Based Approach", this project was cooperation Robotics LAB, Zewail University, Giza, Egypt via prof. Moustafa Elshafei

Journal Reviewer

1. Annals of Telecommunications
2. Journal of Communications (JCM)
3. Journal of Telecommunications and Radio Engineering
4. Journal of Laser Science and Applications

TPC Member

1. IEEE International Conference Novel Intelligent and Leading Emerging Sciences (NILES), Nile University, Egypt
2. IEEE International Japan - Africa Conference on Electronics, Communications & Computations (JAC-ECC), Egypt-Japan University of Science and Technology (E-JUST), Egypt
3. A member of UAVC Judge Board in the 8th International Competition of the Military Technical College (ICMTC 2024)", Military Technical College, Cairo, Egypt
4. A member of UAVC Judge Board in the 7th International Competition of the Military Technical College (ICMTC 2023)", Military Technical College, Cairo, Egypt
5. A member of UAVC Judge Board in the 6th International Competition of the Military Technical College (ICMTC 2022)", Military Technical College, Cairo, Egypt
6. A member of the 1th IEEE Egyptian Paper Contest 2023 Judge Board that organized by IEEE SAC Egypt at Arab Academy for Science, Technology and Maritime Transport, Alexandria, Egypt
7. Session Chairman of the session "Electrical Communications", IAET Scientific Conference Student Activities 2021 (ISC-21)

Scientific Committee Member

1. IEEE Member
2. Counselor of IEEE IAET Student Branch

Technical Programming Languages

1. MATLAB - Professional
2. PYTHON - Medium
3. C/C++ - Beginner
4. LabVIEW - Medium
5. ROS - Beginner

Research Interest & Future Directions

1. Multicarrier Systems (OFDM & FBMC)
2. MIMO-OFDM & MIMO-FBMC Systems
3. Multiple Access Techniques (NOMA Systems)
4. Linear and Nonlinear Equalization Techniques
5. Interference Mitigation and Cancellation
6. Indoor Optical Wireless Communications mainly Visible Light Communications (VLC)
7. Fluid Antenna Systems (FAS)
8. Reconfigurable Intelligent Surface (RIS)
9. Massive MIMO Systems
10. Rate Splitting Multiple Access (RSMA) Technique
11. Deep Learning for Wireless Communication Systems

Publications

Journal Articles

- 2024 **Mohamed S. Elbakry**, Hamed. F. El-Shenawy, Abd El-Hady. A. Ammar, and Basem M. ElHalawany. Ici cancellation over time varying channels via hybrid symbol-time compression and frank-windowing. *Under Review*, 2024.
- 2023 **Mohamed S. Elbakry** and Khaled Ramadan. A time mirroring (tm) scheme for ici mitigation in time varying channel for dvb systems. *Telecommunications and Radio Engineering*, volume 82, pages 1–16. Begel House Inc., 2023.
- 2023 **Mohamed S. Elbakry**, Sherif k. Eldyasti, and Khaled Ramadan. Non-orthogonal algorithm with fourier transform symmetry for performance enhancement of dco-ofdm systems. *Journal of Advanced Research in Applied Sciences and Engineering Technology*. Semarak Ilmu, 2023.
- 2023 **Mohamed S. Elbakry**, Amira A Amer, and Tawfik Ismail. Enhancing performance of downlink noma-based c-ran topology through optimal user pairing and dynamic power allocation scheme. *IEEE Access*. IEEE, 2023.
- 2023 Khaled Ramadan, Basem M, and **Mohamed S. Elbakry**. Performance improvement for dco-ofdm and aco-ofdm systems using symbol time compression. *Telecommunication Systems*, volume 84, pages 77–100. Springer, 2023.
- 2023 Khaled Ramadan, Mohamed A. Abdelhamed, and **Mohamed S. Elbakry**. Nonlinear equalization for performance enhancement of fbmc systems using walsh transform. *IETE Journal of Research*, pages 1–9. Taylor & Francis, 2023.
- 2023 Abdulwahid Mohammed, **Mohamed S. Elbakry**, Hassan Mostafa, and Abdelhady Abdelazim Ammar. Doubling the Number of Connected Devices in Narrow-band Internet of Things while Maintaining System Performance: An STC-based Approach. *Advances in Science, Technology and Engineering Systems*, volume 8, pages 1–10, 2023.
- 2022 **Mohamed S. Elbakry**, Abdulwahid Mohammed, and Tawfik Ismail. Throughput improvement and papr reduction for ofdm-based vlc systems using an integrated stc-imadjs technique. *Optical and Quantum Electronics*, volume 54, pages 1–17. Springer, 2022.
- 2022 Khaled Ramadan and **Mohamed S. Elbakry**. Efficient equalization and carrier frequency offset compensation for underwater wireless communication systems. *Annals of Data Science*, volume 9, pages 1201–1221. Springer, 2022.
- 2020 Mohamed Yasin I Afifi, **Mohamed S. Elbakry**, El-Sayed SA Soliman, and AA Ammar. An efficient technique for out-of-band power reduction for the eliminated cp-stc-shaped system for 5g requirements. *International Journal of Electrical and Computer Engineering*, volume 10, page 5306. IAES Institute of Advanced Engineering and Science, 2020.

Conference Articles

- 2025 **Mohamed S. Elbakry**, Abdelrahman M. Mahmoud, Sherief Hashima, and Basem M ElHalawany. Autonomous navigation for mobile robots in dynamic environments: A comprehensive study using deep learning algorithms. In *2024 Japan-Africa Conference on Electronics, Communications and Computers (JAC-ECC2024)*. IEEE, 2025.
- 2024 **Mohamed S. Elbakry**, Sherief Hashima, and Basem M ElHalawany. Symbol time compression for ofdm spectral efficiency in visible light communication. In *The 17th International Conference on Computer and Electrical Engineering (ICCEE 2024)*. IOP, 2024.
- 2023 **Mohamed S. Elbakry**, Abdelrahman M. Mahmoud, SeifEldeen Hassan, and Tawfik Ismail. Digital twin simulations for connected and automated vehicles: A comprehensive study. In *The 3rd IEEE International Conference on Electronic Engineering (ICEEM-2023)*. IEEE, 2023.
- 2017 **Mohamed S. Elbakry**, Hamed A El-Shenawy, and Abd El-Hady A Ammar. A time inversion and symbol time compression (ti-stc) scheme for ici cancellation in high mobility ofdm systems. In *2017 Japan-Africa Conference on Electronics, Communications and Computers (JAC-ECC)*, pages 82–85. IEEE, 2017.
- 2017 **Mohamed S. Elbakry**, Hamed A El-Shenawy, and Abd El-Hady A Ammar. A symbol time compression for ici reduction in high mobility ofdm systems. In *2017 29th International Conference on Microelectronics (ICM)*, pages 1–4. IEEE, 2017.
- 2015 **Mohamed S. Elbakry**, Hamed F El-Shenawy, and Abd El-Hady A Ammar. C16. improved carrier frequency offset correction for ofdm systems based on dual bandwidth and clustered pilot tones. In *2015 32nd National Radio Science Conference (NRSC)*, pages 229–239. IEEE, 2015.
- 2014 **Mohamed S. Elbakry**, Hamed F Elshenawy, and Abdelhady A Ammar. Improved carrier frequency offset correction for ofdm systems based on type-2 control loop. In *2014 31st National Radio Science Conference (NRSC)*, pages 157–166. IEEE, 2014.

Education

- 2018 PhD, Electrical Engineering Department in Electronics and Electrical Communications, Faculty of Engineering, Al-azhar University, Cairo, Egypt
- PhD Summary I received the PhD degree within 2.6 years. Throughout the thesis, the intercarrier interference (ICI) in OFDM systems caused by high mobility is mitigated using the proposed symbol time compression (STC) scheme. This thesis contributes three published papers.
- 2014 MSc, Electrical Engineering Department in Electronics and Electrical Communications, Faculty of Engineering, Al-azhar University, Cairo, Egypt
- MSc Summary I received the MSc degree within 5 years. In this thesis, the OFDM system are evaluated in different scenarios. Then, the impact of carrier frequency offset (CFO) on OFDM system performance is studied, explained and compensated. This thesis contributes two papers for CFO reduction that are published in NRSC conferences. In 2014, the paper awarded best paper award
- 2005 Bachelor of Electronics & Communications Engineering, IAET, EAA.

- Project title : Water Pump Station Controlled using PLC with SCADA
- Project supervisor: Prof. Hanan Kamal
- Graduation project: Distinction2
- Graduation grade: Excellent with honor 3rd rank over 100 students of a group

Languages

1. Arabic Language : Native Level
2. English Language : B2 Level

Awards

- 2024 Supervisor, **6th Rank**, the unmanned aerial Vehicles Competition (UAVC), the 8th International Competition of the Military Technical College (ICMTC 2024), Military Technical College, Cairo, Egypt
- 2023 Supervisor, **10th Rank** over 520 contestants, the international telecommunications conference ITC-EGYPT 2023, Air Defense College, Alexandria, Egypt
- 2022 Supervisor, **8th Rank** over 56 contestants, " Satellite Universities Project" in Egyptian Space Agency (EgSA), Cairo, Egypt
- 2014 **Best Paper Award** in 31th National Radio Science Conference (NRSC)

Referees

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